

vate to remain competitive. Viasat's acquisition of Inmarsat highlights a significant shift, positioning them as leaders in air and maritime markets. SpaceX's Starlink aims to revolutionise information access, challenging traditional satellite players to update technologies. Iridium maintains a competitive edge with advancements like Certus data services. Globalstar's partnerships, particularly with Apple, focus on cost-effective IoT solutions, while OneWeb targets niche markets, including polar regions and maritime services.

Amazon's Project Kuiper aims to deploy its LEO constellation, potentially redefining the competitive landscape and setting the stage for a rivalry with Starlink. The convergence of satellite and cellular networks is evident in Starlink's recent achievements in text and voice communication, highlighting the challenges and opportunities in bridging mobile phones and orbiting satellites.

Smartphones meets satellite

Currently, satellite connectivity in smartphones is emerging, with users primarily able to send emergency signals rather than make calls to other smartphones. However, future developments could enable two-way messaging and even internet connectivity.

Looking ahead, smartphones could use satellite connectivity for two-way messaging and, eventually, even internet connectivity might be possible with the help of satellites.

Thanks to companies like MediaTek, this technology is becoming more affordable for widespread use in devices.

Satellite connectivity typically requires specialised hardware, such as an antenna capable of transmitting and receiving signals from satellites. Communication satellites act as the medium for sending and receiving

terrestrial signals, enabling devices at both ends to communicate via radio waves.

For example, the Huawei Mate 50 smartphones can send messages by connecting to the Beidou satellites for one-way usage. Apple's iPhone 14 series is equipped with emergency SOS messaging, which can transmit messages via satellites in about 15 minutes.

Both Huawei and Apple have incorporated low Earth orbit (LEO) satellite operations, as LEO satellites are placed near Earth, making them more accessible and frequently used for communications, space, and Earth observations.

The integration of satellite connectivity in smartphones is expanding, offering new possibilities for communication beyond Earth. Satellite-to-cell technology is reshaping mobile connectivity, particularly in regions with limited cellular infrastructure. As mainstream adoption of satellite connectivity in smartphones increases, emergency communication and access to remote areas are significantly improved. With companies like AST Mobile, Lynk, Huawei, and Apple leading the charge, the future of mobile connectivity is poised to be both expansive and transformative, ensuring communication knows no bounds. **EFY**

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you stop
learning,
you start
dying.

—Albert Einstein

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